

Ibaad Muhammad

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EDUCATION

University College London

BSc Computer Science

Sep. 2025 – Jun. 2028

London, UK

British School Dhahran

A Levels: Mathematics A*, Further Mathematics A*, Biology A, Chemistry A

Achieved highest mark in Computer Science AS Level in Eastern Province, Saudi Arabia

Aug. 2023 – Jun. 2025

Khobar, Saudi Arabia

EXPERIENCE

Rolls-Royce - Novel Fission and CO₂ Reduction Project

Oct. 2025 – Jan. 2026

- Selected by Rolls-Royce to lead a team of 5 in developing a High Temperature Gas Reactor model, delegating research tasks and coordinating with project supervisors
- Constructed Python-based simulation models for reactor thermodynamics and power cycle analysis, projecting ~150 tonnes of CO₂ reduced daily and visualising results through programmatic graphing
- Validated model accuracy through automated unit testing and literature comparison, achieving results within 1% of published values

J.P. Morgan - Software Engineering Virtual Experience

Jun. 2024

- Debugged and resolved issues across Python, React, and TypeScript codebases to restore correct rendering and live data flow in a financial web application
- Utilised JPMorgan's open-source Perspective library to build a real-time trading dashboard surfacing live stock price feeds

PROJECTS

Habitly | Python, Flask, JavaScript, React, Tailwind CSS, SQLite, SQLAlchemy, Git

Jan. 2026 – Mar. 2026

- Created a full-stack habit recommendation web app with 3 coursemates using React and Flask, serving personalised activity suggestions based on a user's current state (physical energy, social battery, time available)
- Engineered a Python scoring algorithm exposed via a REST API, incorporating real-time local weather data alongside time of day and recency bias to tailor activity recommendations
- Architected the SQLite schema and implemented secure user authentication with hashed password storage

Bioreactor IoT Control System | C++, JavaScript, Node-RED, JSON, MQTT, Git

Oct. 2025 – Dec. 2025

- Engineered C++ firmware for an ESP32 communication module, serialising sensor telemetry from three bioreactor subsystems into JSON payloads and transmitting to a HiveMQ cloud broker via MQTT
- Built a dual-page Node-RED dashboard in JavaScript with live time-series graphs and gauges for temperature, stirring speed, and pH, alongside setpoint controls and per-subsystem kill switches
- Achieved closed-loop control of temperature, stirring, and pH with target accuracies of $\pm 1^\circ\text{C}$ (25–35°C), ± 10 RPM (100–1200 RPM), and ± 0.2 pH (pH 3–7)

Typocalypse | Unity, C#, Git

Sep. 2025 – Feb. 2026

- Developed a 2D cross-platform arcade game in Unity with C#, applying object-oriented programming principles, where players advance by typing to evade AI-controlled zombies, garnering hundreds of impressions
- Implemented online multiplayer functionality using Unity Netcode for GameObjects, integrating Lobby and Relay services for real-time peer connectivity without direct IP hosting
- Designed all in-game art and sound effects to create a cohesive visual and audio style

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, SQL, HTML, CSS, JavaScript, TypeScript

Frameworks & Libraries: Flask, React, Node.js, NumPy, pandas, Matplotlib, Pygame, SciPy, pytest, SQLite

Developer Tools: Git, Docker, VS Code, PyCharm, IntelliJ IDEA, Claude Code, WSL, Unity, Vercel, Firebase

ACHIEVEMENTS & INTERESTS

- Published several original video games, totalling ~1000 impressions and generating revenue
- Awarded Gold and achieved highest score in school in the UKMT Senior Mathematical Challenge
- Paid Computer Science Outreach Ambassador at UCL, teaching younger students fundamental computing concepts through engaging workshops
- Attained 1300+ Elo rating in online chess, reflecting strong pattern recognition and analytical thinking
- Started for the school basketball A team, showcasing teamwork and discipline
- Solved 250+ LeetCode problems in topics including trees, dynamic programming, graphs and bit manipulation
- Competed in multiple hackathons and game jams, prototyping functional applications under strict time constraints